

COMP 2522 Lesson 7 Lab - Item Basket

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Objective:

You will be modeling a Basket that holds a specific type of Product. For this lab, we'll be making two types of Toys (ActionFigure and JigsawPuzzle).

Put in package `ca.bcit.comp2522.itembasket`

Class Descriptions

Basket<T> (C)

Data Members:

- `numBaskets`: `int`, static, tracks number of Baskets created.
- `inventory`: `List<T>`, initialized in constructor as an `ArrayList`.

Methods:

- ``checkout()``: creates a `Receipt` object.
- ``checkout(final DiscountCode discountCode)``: creates a `Receipt` object and applies a `DiscountCode` to it.

Constructor:

- `Basket(final List<T> items)`: adds all items to inventory.

Receipt (C)

Nested class inside `Basket<T>`.

Methods:

- ``printReceipt()``: void, prints the number of items and each item in the inventory. Also prints the total price of all the items in inventory.
- ``calculateTotalPriceCAD()``: calculates the total price (in CAD) of all the inventory in

Constructor:

- ``Receipt()``: calls `printReceipt()`.

DiscountCode (C)

Nested static class inside `Basket<T>`.

Data Members:

- `promoCode`: `String`, not null or empty.
- `discountedPercentage`: `double`, not over 100%.

Methods:

- `applyDiscount(final double originalPrice)`: applies the discount code and returns the discounted price.

Product (A)

Data Members:

- `productName`: String, not null or empty.
- `productId`: int.
- `priceCAD`: double.

Methods:

- Getters for instance data.

Overrides:

- `toString()`: returns instance data in a nicely formatted String.
- `compareTo()`: compares Products by productId.

Toy (A)

Extends Product.

Data Members:

- `ageRange`: array, only 2 elements, the min and max age, with the first element being the min age and the second element being the max age. The min age should be less than the max age.

Methods:

- `getAgeRange()`: returns a String showing the age range of this Toy. Example output: "Age range: 5 - 12".

Overrides:

- `toString()`: returns instance data nicely formatted in a String. Also calls `toString()` of parent.

ActionFigure (C)

Extends Toy.

Data Members:

- `franchise`: String, not null or empty.

Methods:

- Getters for instance data.

Overrides:

- `toString()`: returns instance data nicely formatted in a String. Also calls `toString()` of parent.

JigsawPuzzle (C)

Extends Toy.

Data Members:

- `totalNumPieces`: int, more than 0 pieces.

Methods:

- Getters for instance data.

Overrides:

- `toString()`: returns instance data nicely formatted in a String. Also calls `toString()` of parent.

Main(C)

Contains the main method that tests the item basket.

Create a List of Products and initialize it as an ArrayList. Add 2 JigsawPuzzles and 2 ActionFigures to this List.

Inside of `Collections.sort()`, pass the List of Products and a new `Comparator<Product>`. For the Comparator, use an anonymous class to override the `compare()` method. Compare two Products by their product name.

Create a Basket that only takes Products using generics. Then, create a `Basket.DiscountCode` object. Print the Basket with and without the `DiscountCode` and see the differences.

Example answer:

[GitHub repo](#)